

Velora Studio

OpenAI Job + How OpenAI Tells Photoroom

Studio Shot pipeline — Creative Director, prompts, and Photoroom API fields

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Deploy: <https://velora-studio-six.vercel.app>

1. Simple Idea

OpenAI is the brain. Photoroom is the camera studio. WhatsApp is the shop counter.

- Merchant sends a product photo on WhatsApp and answers style / angle / lighting / quality.
- OpenAI looks at the photo, understands the product, and writes technical instructions.
- Velora converts those instructions into Photoroom API form fields + a background scene prompt.
- Photoroom removes background, centers the product, adds scene/shadows, and returns a PNG.
- Velora sends the finished images back on WhatsApp (A !' B !' C) and charges 1 studio credit.

2. Who Does What

OpenAI (GPT-4o / gpt-4o-mini)

- Sees the merchant photo (Vision).
- Decides if the photo is usable.
- Detects product type, open box vs compact, left/right alignment in source.
- Writes `api_blueprint` (padding, shadows, canvas color, uncrop, etc.).
- Writes `lifestyle background.prompt` text (table, props, lighting feel).
- Helps recommend style / angle / lighting during the WhatsApp interview.

Photoroom (Image Editing API v2/edit)

- Cuts out the product (`removeBackground`).
- Places it dead-center with locked padding margins.
- Renders white catalog background OR AI lifestyle scene from `background.prompt`.
- Applies shadows, lighting, text removal, beautify when asked.
- Exports PNG (usually 1000×1000).

Velora engine (our code)

- Asks WhatsApp questions and stores user choices.
- Calls OpenAI, then maps JSON !' multipart Photoroom fields.
- Locks safety rules (center, padding 0.18/0.22) so model mistakes cannot break framing.
- Retries safer Photoroom payloads if an advanced field fails.

3. Studio Shot Chain (Step by Step)

- 1) Photo arrives on WhatsApp !' language (if needed) !' Mode: Studio shot.
- 2) OpenAI Call #1 — Product analysis (Vision). Used for WhatsApp intro + AI Recommended style/angle/lighting. Does NOT talk to Photoroom yet.
- 3) WhatsApp interview: Style !' Angle !' Lighting !' Quality (diecut / AI recommended can skip some steps).
- 4) Generation starts. For each output image, OpenAI Call #2 — Creative Director (Vision) returns `api_blueprint` JSON.
- 5) For lifestyle images only: OpenAI Call #3 — Background prompt writer turns vibe (style + angle + lighting + quality) into one Photoroom `background.prompt` paragraph.

6) Engine builds Photoroom multipart form from blueprint + prompt !' POST /v2/edit !' PNG !' WhatsApp.

4. OpenAI Call #1 — Interview Analysis

When: Right after merchant chooses Studio shot, before style list.

Job: Understand the product so WhatsApp can recommend options.

- Model: gpt-4o (Vision + JSON)
- Returns: category, summary, recommended style/angle/lighting, photo issues, isolateFirst, etc.
- Goes to: WhatsApp copy + recommendation lists — NOT Photoroom API fields.

5. OpenAI Call #2 — Creative Director (Brain for Photoroom)

When: Every catalog or lifestyle render (per image). This is the main way OpenAI “tells” Photoroom what to do.

Job: Look at the photo like a commercial photographer and output a technical blueprint.

- Model: gpt-4o (Vision + strict JSON)
- Detects: silhouette (compact vs tall/open gift box), source_alignment (left/center/right), shadows, canvas color, uncrop, pre-cutout, beautify, relight.
- Engine then LOCKS padding: compact !' 0.18 | open/tall !' 0.22
- Engine ALWAYS forces output center — even if source photo was left-aligned.

Example api_blueprint shape OpenAI returns:

```
{
  "detected_category": "jewellery_accessories",
  "product_name": "Open gift box",
  "is_usable": true,
  "user_guidance": "Looks good! Studio shot ready.",
  "logo_safety_note": "Keep brand text sharp",
  "api_blueprint": {
    "output_size": "1000x1000",
    "padding": 0.22,
    "silhouette": "tall_open_asymmetrical",
    "source_alignment": "left",
    "subject_pose": "upright",
    "shadow_direction": "behindLeft",
    "shadow_intensity": 0.5,
    "shadow_softness": 0.8,
    "canvas_bg_color": "FFFFFF",
    "enable_beautify": false,
    "enable_relighting": true,
    "requires_uncrop": false,
    "requires_pre_cutout": false,
    "export_format": "png"
  }
}
```

Note: User angle / lighting / quality choices are NOT inside this blueprint. Placement and shadows come from Creative Director + engine locks. Angle/lighting/quality mainly shape the lifestyle scene text (Call #3).

6. OpenAI Call #3 — Lifestyle Background Prompt

When: Lifestyle / ad-style images only (not solid white catalog packshot).

Job: Write ONE paragraph for Photoroom background.prompt — surface, exact props, lighting, centered product feel.

- Model: gpt-4o / helper text model
- Input vibe includes: user style label + angle/lighting/quality prompt suffixes + product context

- Rules: exact prop counts (e.g. two diyas), no abstract yellow clutter, product planted on a table
- Sent to Potoroom as: background.prompt (+ expandPrompt.mode = ai.auto)

7. How OpenAI Instructions Become Potoroom Fields

OpenAI never calls Potoroom directly. Velora maps the blueprint JSON into multipart form-data for POST <https://image-api.potoroom.com/v2/edit>

Always sent (placement / safety):

- removeBackground = true
- referenceBox = subjectBox (use cutout box, not original left/right photo framing)
- ignorePaddingAndSnapOnCroppedSides = false (do not stick product to left edge)
- horizontalAlignment = center
- verticalAlignment = center
- padding = 0.18 or 0.22 (locked)
- position.mode = custom | position.padding | position.*Alignment = center
- outputSize = 1000x1000 | export.format = png
- textRemoval.mode = ai.artificial

From Creative Director blueprint:

- catalog mode !' background.color = canvas_bg_color (e. and g. FFFFFFFF / F5F5F5)
- lifestyle mode !' background.prompt = OpenAI scene paragraph
- enable_relighting !' lighting.mode = ai.auto
- enable_beautify !' beautify.mode = ai.auto
- requires_uncrop !' uncrop.mode = ai.auto
- shadow.* overrides from shadow_direction / intensity / softness / subject_pose

If Potoroom rejects advanced fields, engine retries: drop textRemoval !' drop shadows !' minimal centered payload.

8. Catalog vs Lifestyle vs Diecut

- White studio / ecommerce: Image A = catalog (solid color from OpenAI canvas_bg_color). Images B & C = lifestyle (background.prompt from OpenAI).
- Other scene styles: up to 3 lifestyle images, each with its own OpenAI Creative Director + background prompt.
- Diecut / transparent PNG: Potoroom cutout only — no Creative Director blueprint for scene.

9. What User Answers Affect

WhatsApp choices vs Potoroom:

- Language !' UI text only
- Style !' catalog vs lifestyle vs diecut + scene vibe
- Angle / Lighting / Quality !' scene vibe text (then rewritten by OpenAI Call #3) — NOT Potoroom center/padding fields
- Creative Director (automatic) !' Potoroom padding, center, shadows, canvas, uncrop, beautify

10. One-Line Summary

OpenAI looks at the product photo and writes a technical shot list (api_blueprint) plus an optional scene description; Velora translates that into Potoroom form fields; Potoroom renders the studio PNG; WhatsApp delivers it to the merchant.

Key code: web/src/lib/studio-analysis.ts | studio-analysis-prompt.ts | studio-engine-core.ts | potoroom.ts | studio-

whatsapp.ts

File: docs/Velora_OpenAI_Photoroom_Guide.pdf